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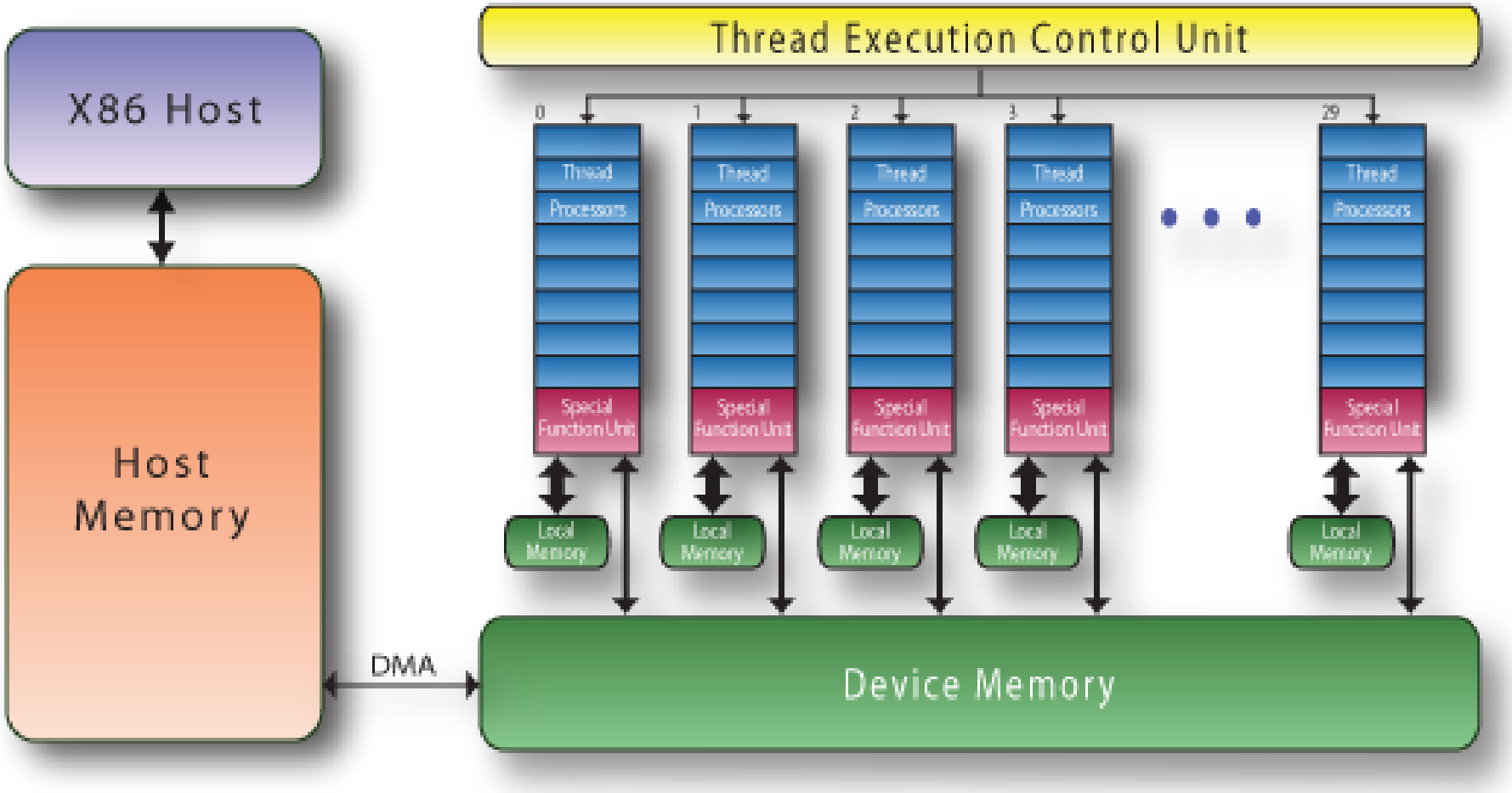
High Performance Computing

GPU

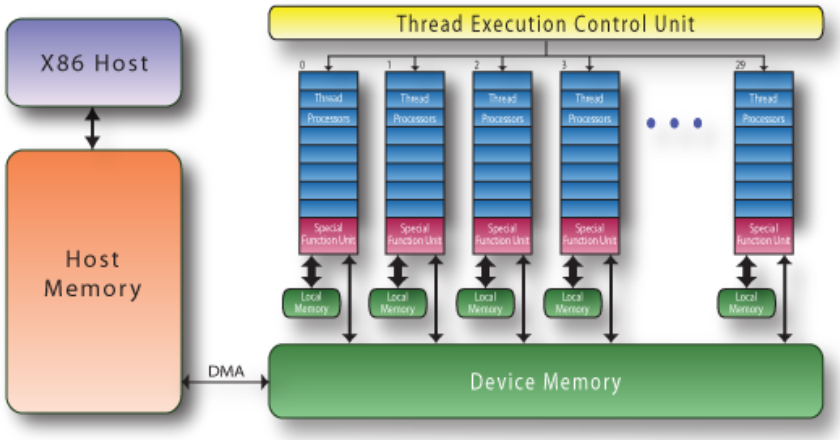
Dr Ligang He

Architecture of GPU



GPU - Graphics Processing Unit

Specification	RTX 3090	RTX 4090	RTX 5090
SM Count	82	128	170
Cores per SM	128	128	128
Total Cores	10,496	16,384	21,760
Release Date	September 2020	October 2022	January 2025



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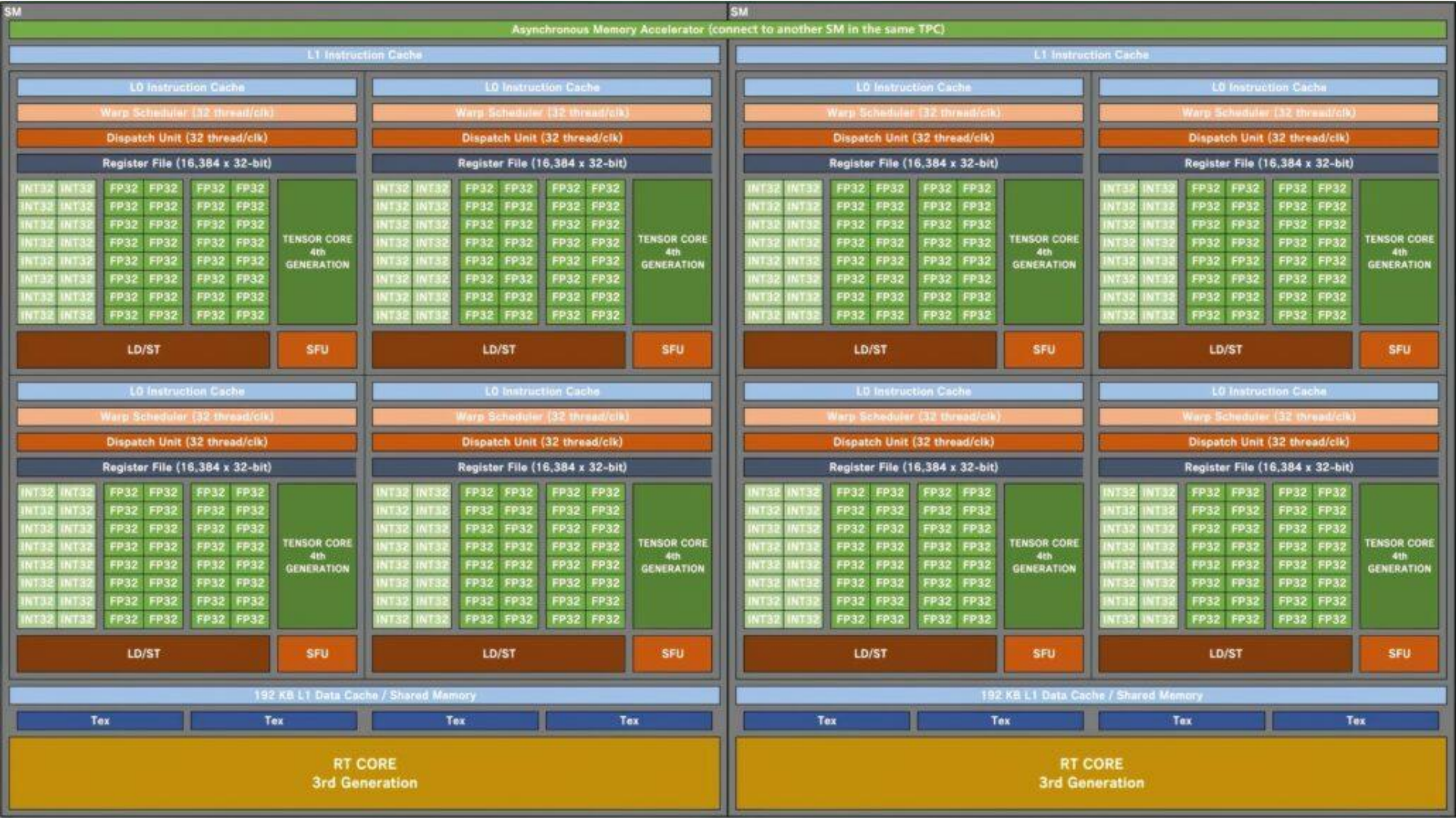
RTX 3090 has the Ampere architecture

SM Internal of Ampere architecture



⑩ Ada Lovelace architecture: RTX 4090

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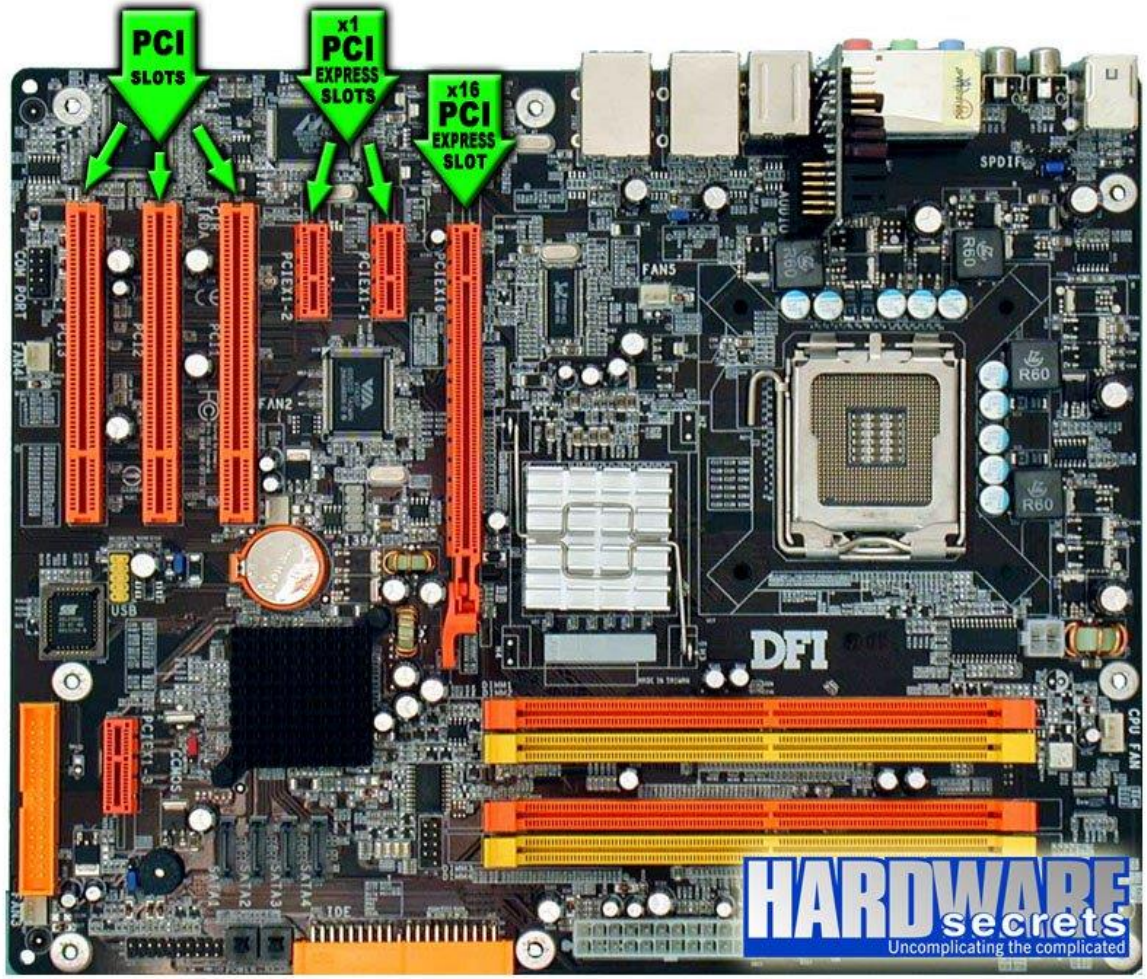
Hopper architecture (H100)

GPU is a PCI-e peripheral device

- PCI-e slot



Performance Trend

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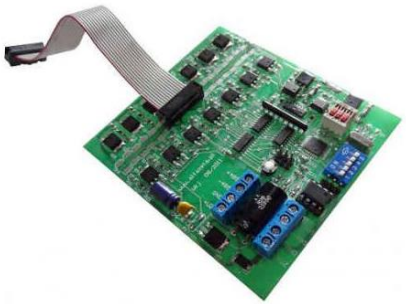
- GPU is typically more powerful than multicore CPU by an order of magnitude

- Why is there such performance gap?
  - Because of the differences in the design between GPU and CPU

-

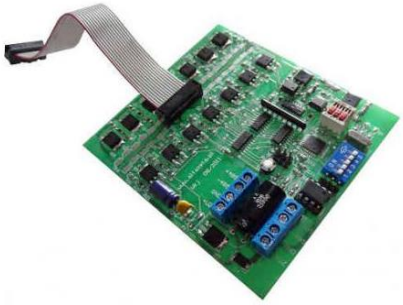
Design of CPU

- **The design objective of CPU is to optimize the performance of a sequential code**
  - **Control unit is important in CPU**
    - Obtain instructions from memory
    - Interpret the instructions
    - Figure out what data are needed and where they are stored
    - Signal the functional units (ALUs) to run the instructions



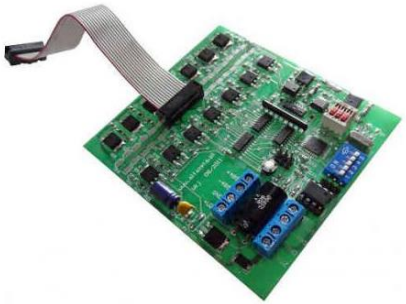
Design of CPU

- The design objective of CPU is to optimize the performance of a sequential code
  - Control unit is important in CPU
  - CPU has complicated control units: a program may contain various types of instructions and logic
    - Execute instructions out of their sequential order (single core) or in parallel (multicore)
    - Branch prediction
    - Data forwarding



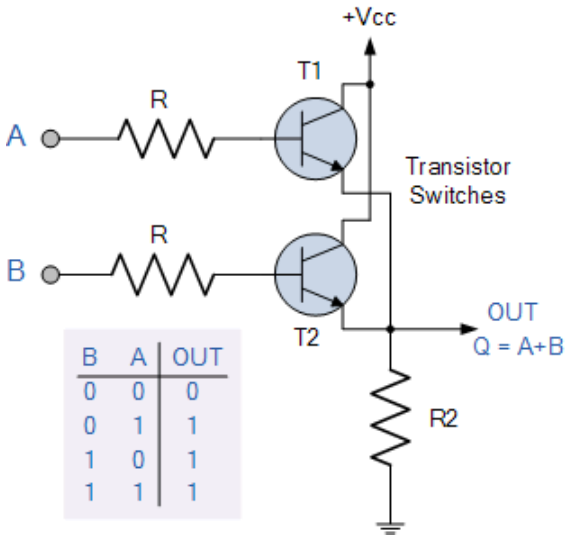
Design of CPU

- The design objective of CPU is to optimize the performance of a sequential code
  - Has complicated control unit
  - Has large cache to reduce data access latencies
  - Powerful ALU



Design Objective of CPU

- **Latency-oriented design:** minimize the average latency of instructions in a sequential code
  - Complicated control unit
  - Large caches on CPU chip (on-chip cache)
  - Powerful ALU
  - They are at the cost of increased use of chip area and power
- Applications with one or very few threads achieve higher performance in CPU



OR gate with transistors

Motivation of GPU Design

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- Video game industry: need to perform a massive number of floating-point and integer calculations per video frame

- Motivate GPU vendors to use most chip area (transistors) for floating-point/integer calculations
  - Each calculation is simple
    - Simple control unit and simple ALUs
  - Calculation is more important than cache
    - Small cache, allowing data access to have long latency

GPU Design

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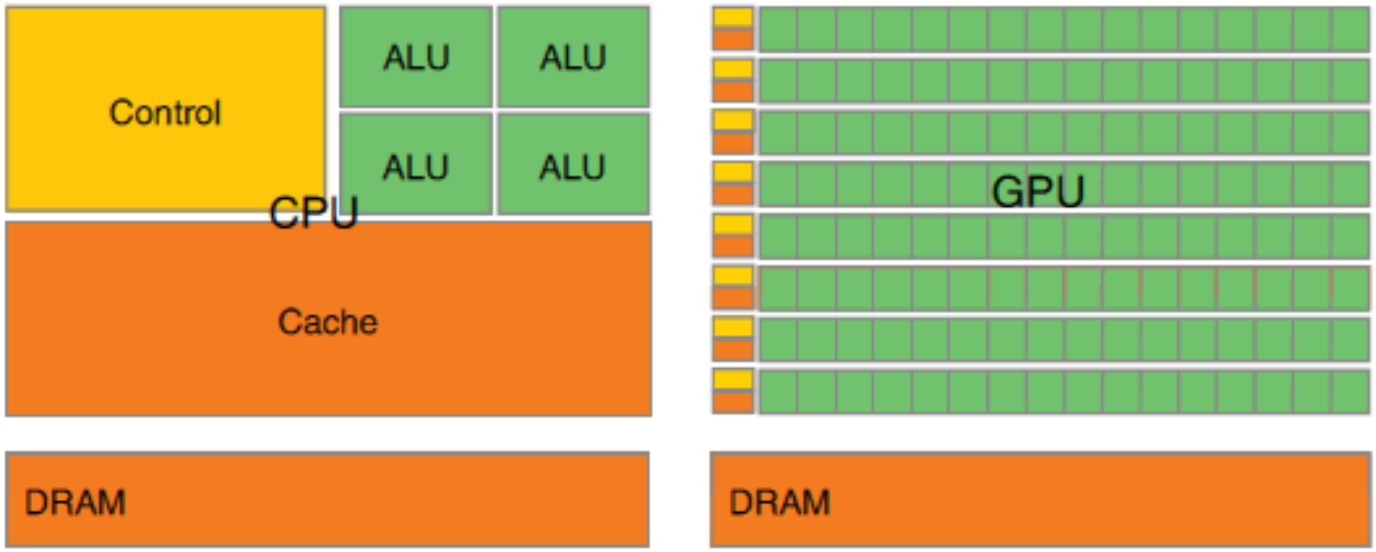
- Use a large number of simple ALUs to increase the total throughput

- Throughput is the number of instructions completed in a time unit
- The application runs with a large number of threads
- While some threads are waiting for long-latency operations (e.g., memory access), GPU can always find other threads to run
- Throughput-oriented design: maximize the total throughput of a large number of threads, allowing individual threads to take longer time

GPU vs. CPU in Architecture